

ual:

Coding Three

Creative Robotics

Week 1: 2 October 2025

Today's Agenda

Welcome to Year 2!

Unit overview

Course admin (SEATS, Slack, GitHub)

Introduction to Creative Robotics and
Computer Vision

Lab: Assignment 1 notebooks



Welcome + Introduction



Introduction

- Welcome to Year 2!
- Please tell us a little bit about yourself: Your name, pronoun and one thing about you people wouldn't be able to guess by looking at you
- What interests you most with machine learning and AI?
- Is there any artistic/media forms that you want to combine with robots (sculpture, performance, installation, sound, projection, film, etc.)?

Class Structure

- In this course, you will learn the foundations of machine learning and AI so that you can apply these tools in your creative robotics work
- Each class is structured with two lectures with two breaks, plus lab time, where you can ask questions, work on projects and the five in-class assignments

Learning Outcomes

LO1: Understand how machine learning works in practice (Knowledge)

LO2: Demonstrate practical skills in computer programming for Computer Vision contexts (Process)

LO3: Use existing frameworks and examples to create computational systems that make use of Computer Vision approaches relevant to robotics (Realisation)

Introduction + Class Structure

- Week 1-3: Foundations and Basic Computer Vision
 - OpenCV basics: erosion, dilation, blob detection
 - Background subtraction, edge detection
 - Contour detection and tracking
 - Machine learning fundamentals

Introduction + Class Structure

- Week 4-7: Machine Learning Implementation
 - Keras and TensorFlow
 - Neural networks
 - Training, loss functions, back propagation
 - CNNs and image classifications

Introduction + Class Structure

- Week 8-10: Advanced ML & Project Development
 - Pre-trained models and transfer learning
 - Feature visualization
 - Activity detection, DETR, VQA
 - Tutorial sessions on project development (both Elements)
 - *JoJo will sub for me during Week 10 and 11 (4 and 11 December)
- Week 11-13: Assessment Prep and Submissions

Assessment

- Element 1: In-class Assignments (50%)
 - There will be five throughout the term
- Element 2: Mini Project (50%)
 - Creative project applying computer vision to robotics

We will discuss these more in details next week.

Class Admin: SEATS

■ Please download the new SEATS app if you haven't done so



Apple App Store



Google Play

Class Admin: Slack Access

Slack access for all CCI students:

- ★ Please follow this QR code to log into Slack with your UAL username and password
- ★ If you're having trouble with your UAL login, please make sure you've changed your password before your first login via Password Self Service:
<https://pss.arts.ac.uk>
- ★ Remember to install the iPhone/Android app and Desktop app for your computer!
- ★ Once you've joined create your profile, don't forget to add a nice image of yourself :)



Getting Started with GitHub

- All final submissions will be via GitHub
 - This means you will need your own GitHub account
 - Use your UAL credential to gain access
- Your lab notebooks will also be delivered via a GitHub Repository.
- Set up your own account first by going to git.arts.ac.uk

Here is the handy guide from CCI Wiki.

It is also recommended to use IDE tools like Visual Studio Code to work on your notebooks.

Resources

- There are a lot of support resources both within and outside of CCI
- Within CCI, your colleagues, your course tutor (Cindy), course leaders, technicians (Mayra, based in Greencoat, and Ilia, based in High Holborn, both specialize in AI and machine learning)
- Outside of CCI:
 - Community help: Stack Overflow, Hugging Face Forums, GitHub Discussions/Issues (specific coding questions, model help, and project-specific troubleshooting)
 - Official documentation: Python (docs.python.org), NumPy (numpy.org), OpenCV (docs.opencv.org/4.x/), TensorFlow/PyTorch docs — consult these first for API details and correct usage.
 - Research & implementations: Google Scholar, Papers with Code (paperswithcode.com), conference publications: use for finding papers, code implementations, references, etc.

Class Slack Channel #cr-coding3-25

- We have a Slack channel for the class: #cr-coding3-25
- This channel is for any technical question you have related to the course. Please default to asking questions in the public channel ❤️
 - Someone else may have the same question as you, or your colleague can also jump in to help
- If you have any sensitive/private questions, please DM Cindy on Slack or email
- Any admin-related questions, please contact the CCI Admin team
- Book tutorials with the technicians here

Technical Skills Workshops

■ Highly recommend!

TECH SKILLS PROGRAMME AUTUMN 2025

Oct 1st
GB_G03 → 3:00 - 5:00pm → Intro to command line → Agnes
HH_302 → 4:00 - 6:00pm → Python Jumpstart → Mayra and Ilia

Oct 8th
PR courtyard → 3:00 - 5:00pm → Fix your bike → Technical team

Oct 15th
GB_G03 → 3:00 - 5:00pm → 3D Scanning → Lieven
HH_302 → 4:00 - 6:00pm → Git and Github → Agnes

Oct 22nd
GB_G03 → 3:00 - 5:00pm → Python Jumpstart → Mayra and Ilia

Oct 29th
GB_G03 → 3:00 - 5:00pm → 3D sprinting → Yuwei
HH_302 → 4:00 - 6:00pm → Python data manipulation → Mayra

Nov 5th
GB_G03 → 3:00 - 5:00pm → Laser laser laser → Yuwei

Nov 12th
GB_G03 → 3:00 - 5:00pm → eTextiles → Agnes and Rosie
HH_302 → 4:00 - 6:00pm → Intro to jetson nano → Ilia

Nov 19th
GB_G03 → 3:00 - 5:00pm → Electronics fabrication → Agnes

Nov 26th
GB_G03 → 3:00 - 5:00pm → Debugging code → Agnes and Lieven
HH_302 → 4:00 - 6:00pm → Max MSP → Tom Webster

These workshops require no prior experience and are open to students and staff across all CCI campuses. Unless specified, no sign up is required, just turn up.

GB → Greencoat
PR → Peckham Road
HH → High Holborn

more info → cci.to/tech-skills





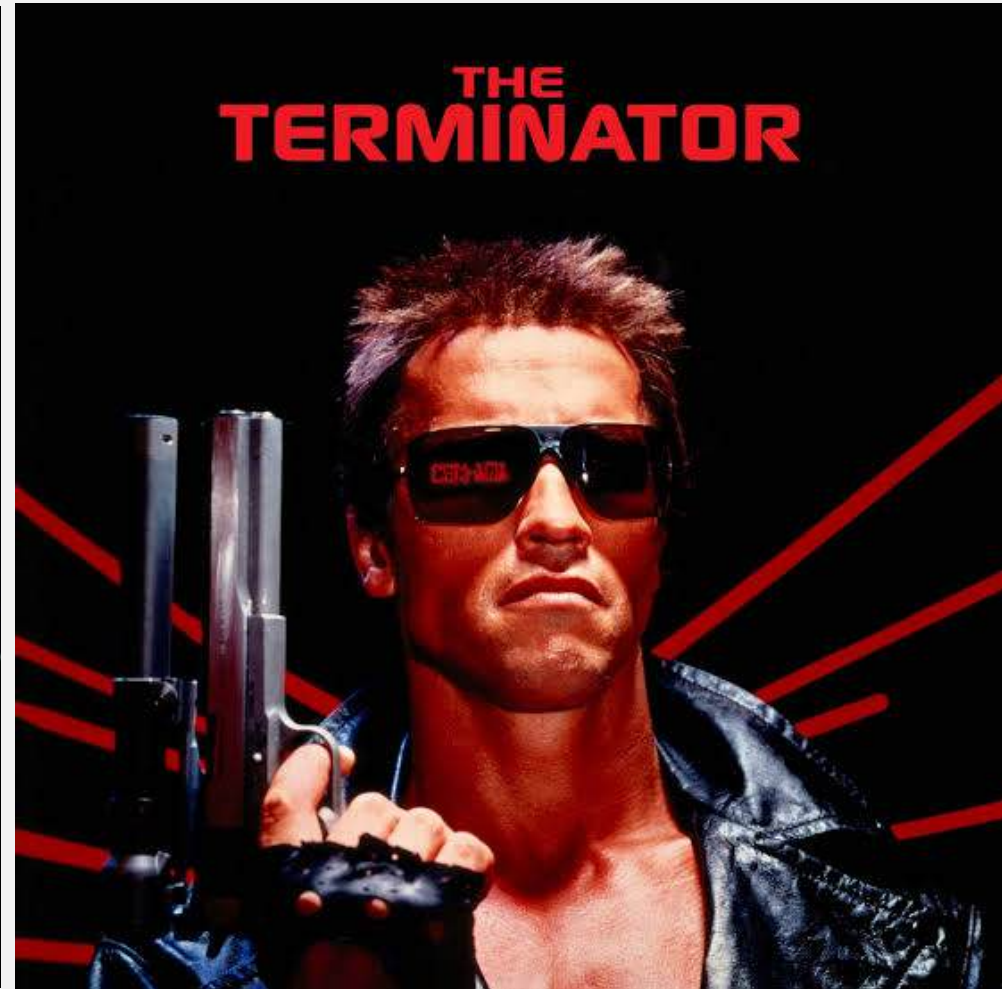
Robots x Machine Learning and AI



When people say Robots with AI...

■ What do you think of when you hear “AI and Machine Learning” in Robotics?

When people say Robots with AI...



But in reality...



ual:

What is Machine Learning and AI?

AI and Machine Learning:

- **Aim to** (meaning goals, **NOT REALITY**):
 - Simulate cognitive processes (perception, reasoning, problem-solving, decision-making)
 - Learn from data, patterns and experiences
 - Develop ability to adapt to new contexts
 - Functionality can be developed without human explicitly creates step-by-step programming to do so

AI and Machine Learning:

■ Current reality:

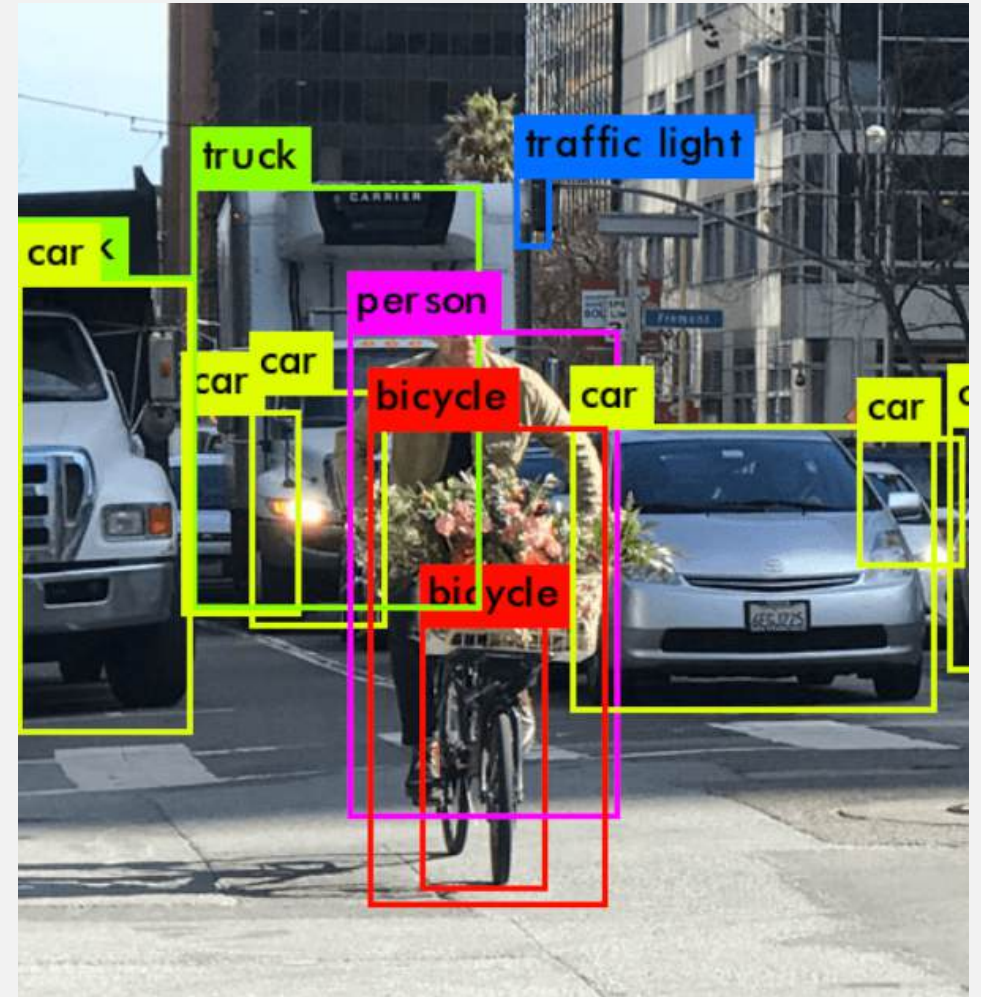
- We have narrow, task-specific AI (image recognition, game playing, language processing) and they are far from perfect still
- Still far from general human-like intelligence. AGI (Artificial General Intelligence) is still theoretical
- Good at specific tasks that it is trained for, but can struggle/ hallucinate with anything outside of the training data

What is Computer Vision?

- Computer vision is a subfield of AI/ML focusing on visual perception
- Methods for acquiring, processing, analyzing and understanding digital images
- Extraction of high-dimensional data from the real world
- With the objectives of producing numerical or symbolic information, e.g. in the forms of decisions
- Enables computers to “understand” images

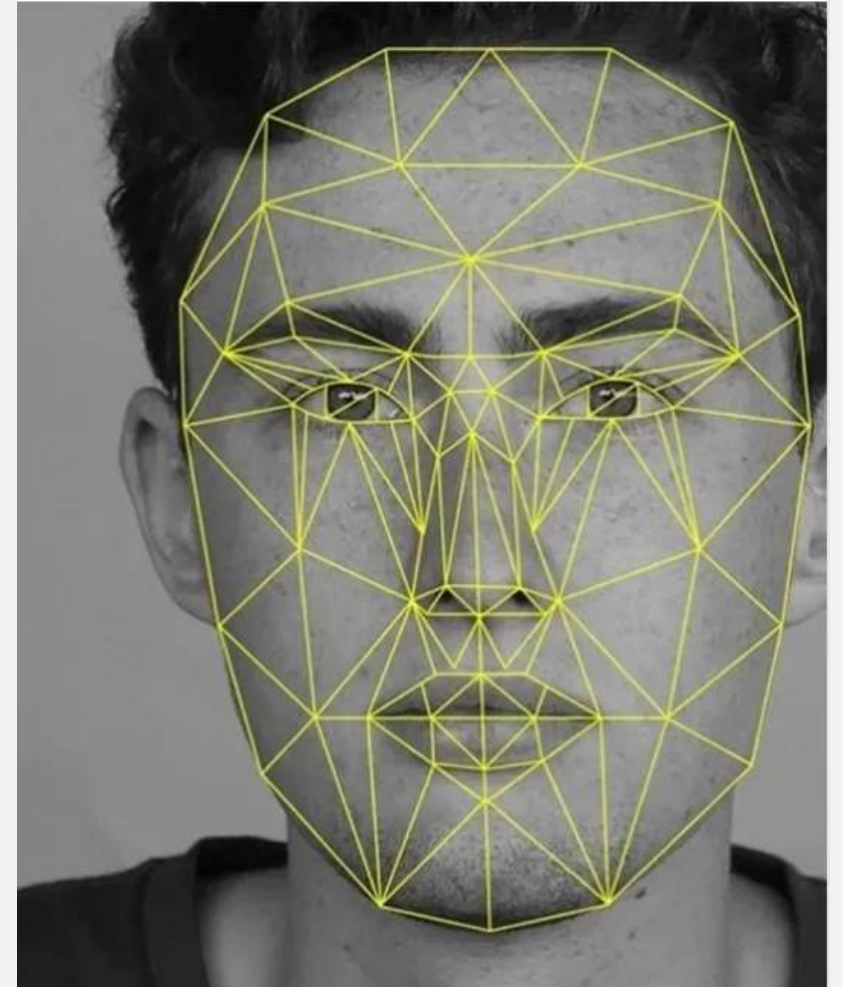
Can Do: Object Detection and Recognition

- Identify and locate specific objects within an image
- Label what the computer "sees" (e.g., person, car, dog, chair)
- Draw bounding boxes around detected objects
- Assign confidence scores to detections
- Real-world applications: Self-driving cars, inventory management, security systems



Can Do: Face Detection and Facial Recognition

- Detect presence and location of human faces in images
- Identify specific individuals (facial recognition)
- Analyze facial features and landmarks
- Detect facial expressions and emotions
- Real-world applications: Security systems, social media tagging, interactive installations, emotion-aware interfaces



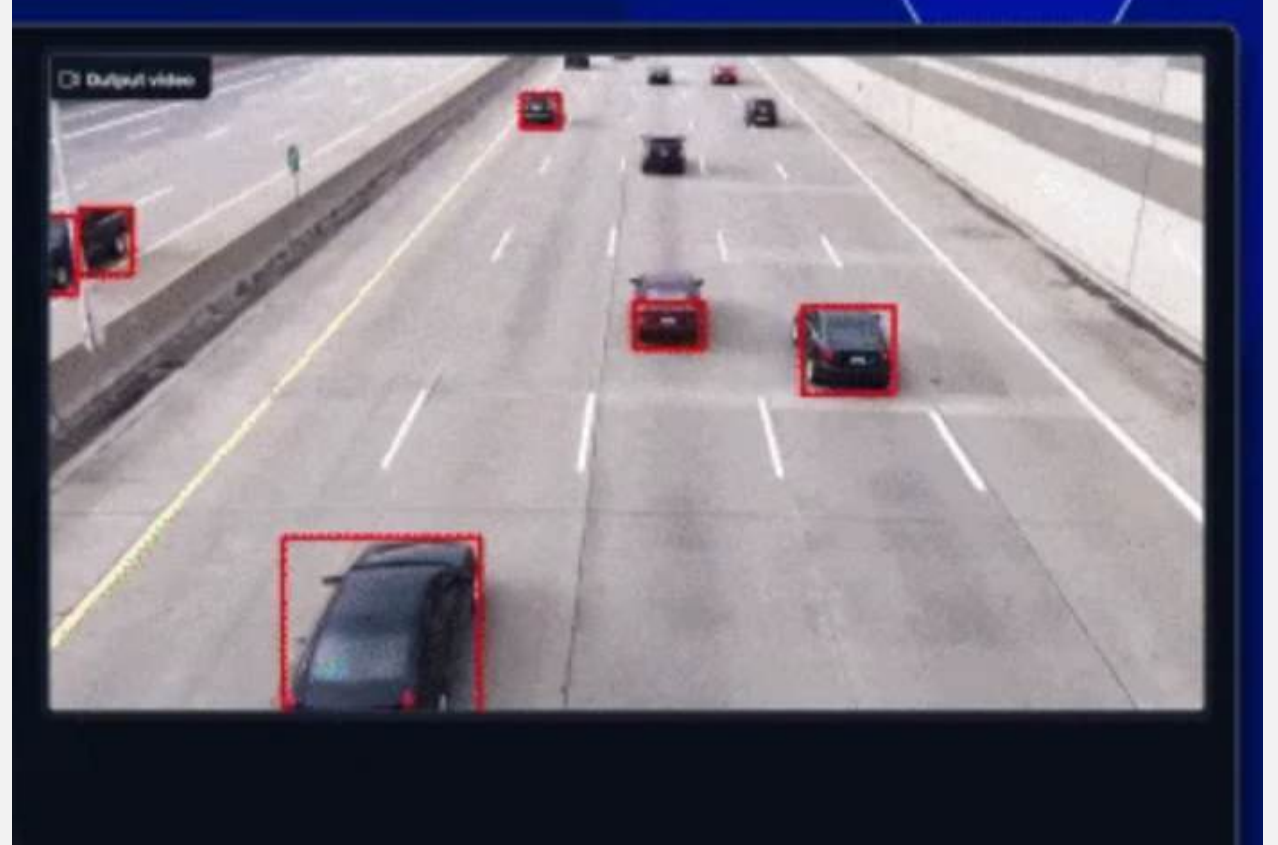
Can Do: Pose and Gesture Tracking

- Detect human body positions and skeleton structure
- Track movement of body parts over time
- Recognize specific gestures and actions
- Enable body-based interaction without physical controllers
- Real-world applications: Motion capture, fitness apps, interactive art, gesture control interfaces



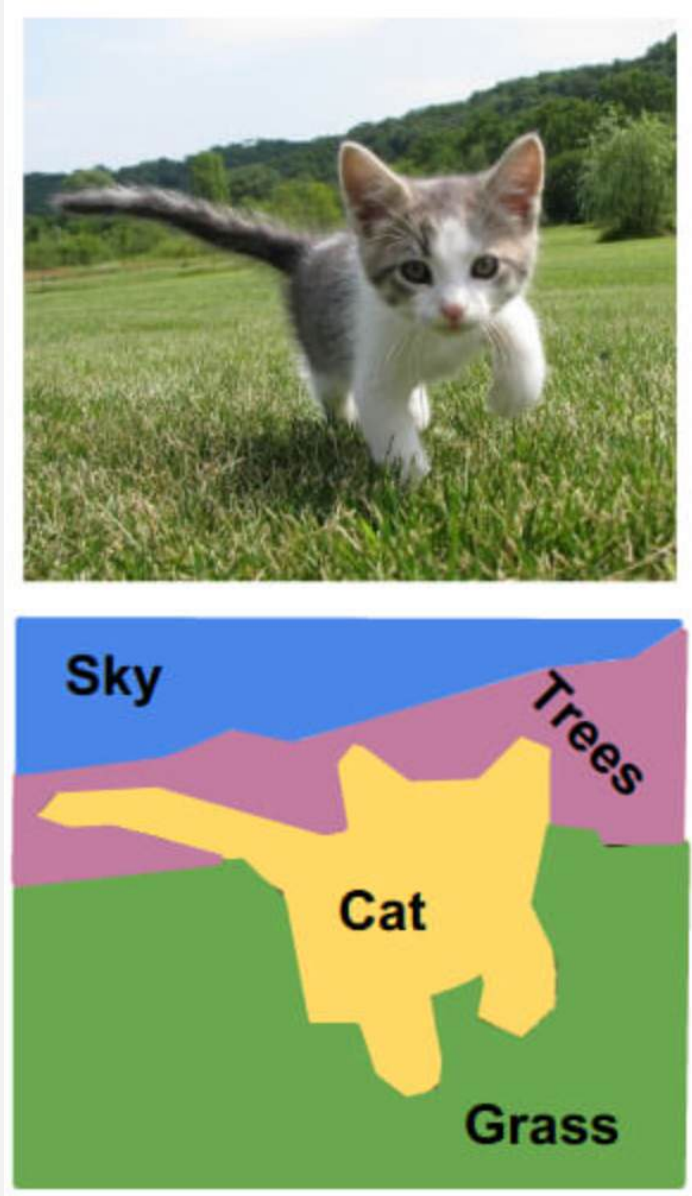
Can Do: Motion Detection & Tracking

- Identify moving objects in video feeds
- Track objects across frames
- Detect changes in scenes over time
- Follow specific targets as they move
- Real-world applications: Surveillance, sports analysis, autonomous robots, interactive installations



Can Do: Scene Understanding and Segmentation

- Classify entire scenes (indoor/outdoor, beach/forest, etc.)
- Segment images into meaningful regions
- Understand spatial relationships between objects
- Interpret context and environment
- Real-world applications: Autonomous navigation, augmented reality, environmental monitoring, image editing tools



Computer Vision in This Course

- These are the kinds of visual understanding tasks we'll learn to implement
 - Start with fundamentals (Week 1-3): Basic OpenCV operations
 - Progress to detection and tracking (Week 3-4)
 - Then machine learning for classification (Week 4-7)
 - Advanced topics (Week 8-10): Pre-trained models, feature visualization

What are Robots?

- Robots are systems that:
 - Carry out tasks autonomously or with minimal human intervention
 - Are programmable: able to sense environment, make decisions and take actions to achieve objectives
 - Equipped with sensors, actuators and control systems that allow them to perceive and interact with robots' environment
 - For example, Roomba vacuums

Sense-Plan-Act

Sense-plan-act" is a fundamental concept in robotics that refers to the three main stages of an autonomous system's operation. In this framework, the robot first senses or perceives the environment using various sensors, such as cameras, microphones, or touch sensors, to gather relevant information about its surroundings. This sensory input is then processed and analysed to generate a plan or a set of actions that the robot should undertake to achieve its goals or respond to the given situation.

Samani, H. (2023) Creative Robotics, p50

Sense-Plan-Act

- SENSE: Perceive environment using sensors (cameras, microphones, touch sensors)
- PLAN: Process sensory input and generate action plan
- ACT: Execute actions to achieve goals or respond to situation
- This is a common pipeline: sensor gathers data, a planner decides what to do (processing) and then actuators execute
- Many systems are closed-loop and iterative

Sense-Plan-Act

- Sense = see/hear/measure
- Plan = decide what to do next
- Act = move or change output
- Robots usually repeat this cycle many times per second or minute

Sense-Plan-Act

■ Reactive Design (Sense → Act):

- These are obstacle-avoidance robot. The bump sensor or distance sensor immediately turns the robot away.
- No planning step here.
- When to pick it: Choose a reactive design when speed and simplicity matter: low-latency responses to immediate sensor inputs are more important than long-term planning.
- Real-life example: A Roomba vacuum that uses bump sensors and short-range IR/sonar to avoid obstacles and change direction immediately.

Sense-Plan-Act

- Sense→Plan→Act classic:

- Navigation with a map:

- Sense: camera/LiDAR senses environment

- Plan: planner computes a path

- Act: motors follow the path

- When to pick it: When the task requires reasoning about future states, long-range goals, or safe paths through a complex environment

- Real-life example: Indoor delivery robots

Sense-Plan-Act

- Learned policy (Sense→Act with implicit planning):
- A neural network mapping camera frames to steering commands
- It appears to jump straight from sensing to action, but the decision-making is encoded in the model
- When to pick it: Pick a learned policy when the decision-making can be learned from data and you need the system to handle complex perceptual inputs end-to-end, but you accept that the policy may be opaque and brittle outside its training distribution
- Real-life example: End-to-end neural driving demos (e.g., NVIDIA's PilotNet research) where camera frames map directly to steering commands via a trained neural network

ML x Creative Robots

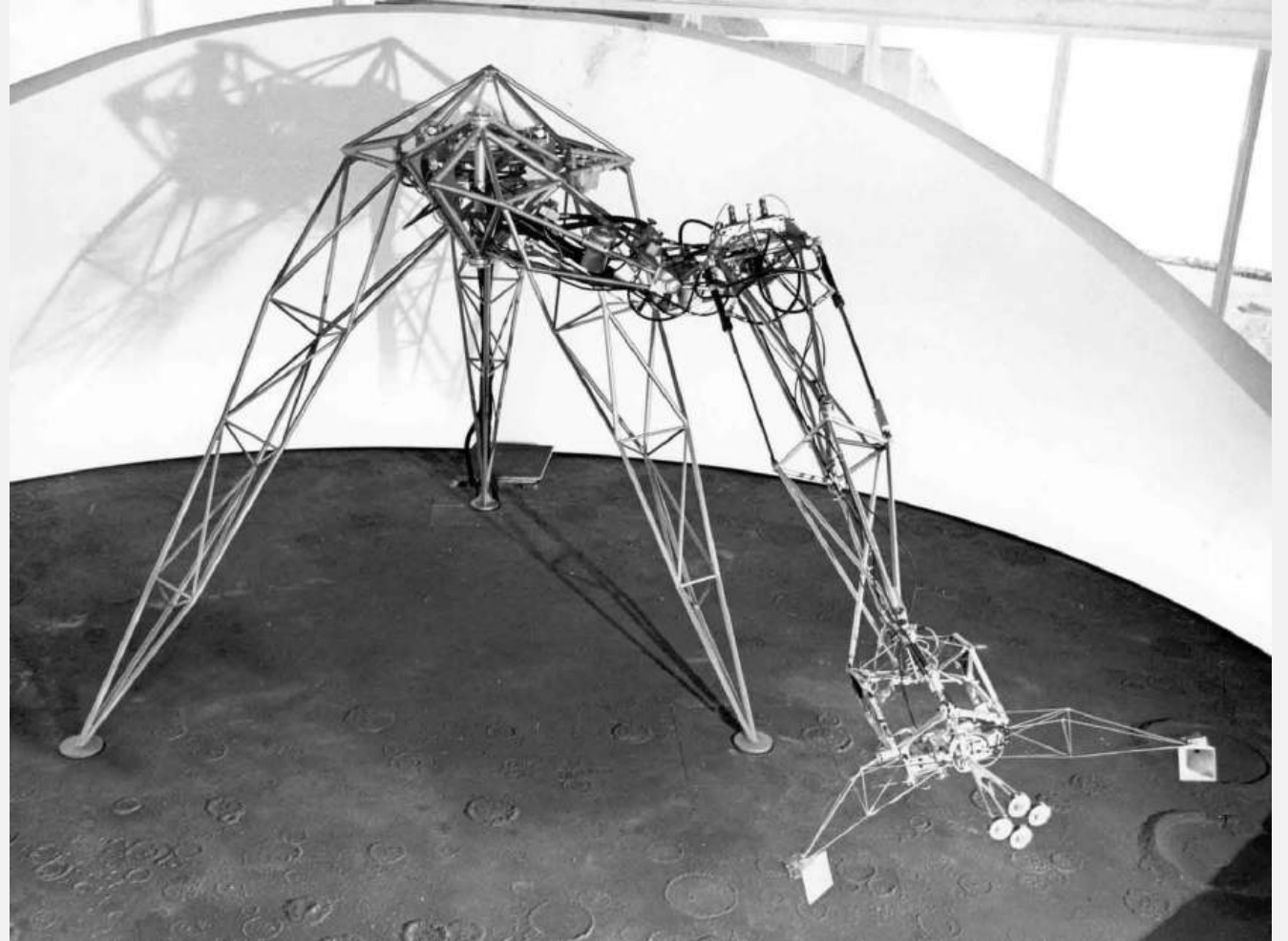
- By combining computer vision with other AI techniques:
 - Robots can engage with audiences
 - Interpret artistic cues and human behavior
 - Create interactive and immersive experiences

ML x Creative Robots

- Computer vision allows robots to:
 - Perceive and understand visual stimuli
 - Recognize people, objects and actions
 - Respond to gestures and movements
 - Collaborate with human artists

Creative Robotics Art Historical Context

- Edward Ihnatowicz, *The Senster*, 1970
- Considered the first work or robotic sculpture to be controlled by a digital computer
- Responded to sound and movement
- **This did not use computer vision.
- It used microphones
- Video: <https://www.youtube.com/watch?v=wY85GrYGnyw>
- Reading



Creative Robotics Art Historical Context

- Simon Penny, *Petit Mal*, 1993
- Interactive robotic installation
- Tracked and followed viewers
- Early exploration of human-robot interaction in art context
- Also *not* computer vision based. It used sensors to detect people's positions and movements
- Video link: <https://simonpenny.net/works/petitmal.html>
- These early works established foundation for field



Creative Robotics Art Historical Context

The goal of Petit Mal was to produce a robotic artwork which is truly autonomous; which was nimble and had “charm;” that sensed and explored architectural space and that pursued and reacted to people; that gave the impression of intelligence and had behavior which was neither anthropomorphic nor zoomorphic, but which was unique to its physical and electronic nature. It was not my intention to build an artificially intelligent device, but to build a device which gave the impression of being sentient, while employing the absolute minimum of mechanical hardware, sensors, code and computational power. My focus was on the robot as an actor in social space. — Simon Penny



Creative Robotics Art Historical Context

Before computer vision, robots like *Petit Mal* used simple sensors.

Computer vision enables robots to understand context, recognize individuals, and respond to complex human behavior—opening entirely new creative possibilities.



Design Philosophy in Creative Robotics

- **Key question: What makes a robot compelling?**
- **Technology vs Behavior:**
 - Is it about the most advanced sensors?
 - Sometimes simple sensing and clever behavior = rich interaction
 - The goal for *Petit Mal* was to create impression of intelligence, autonomy and personality

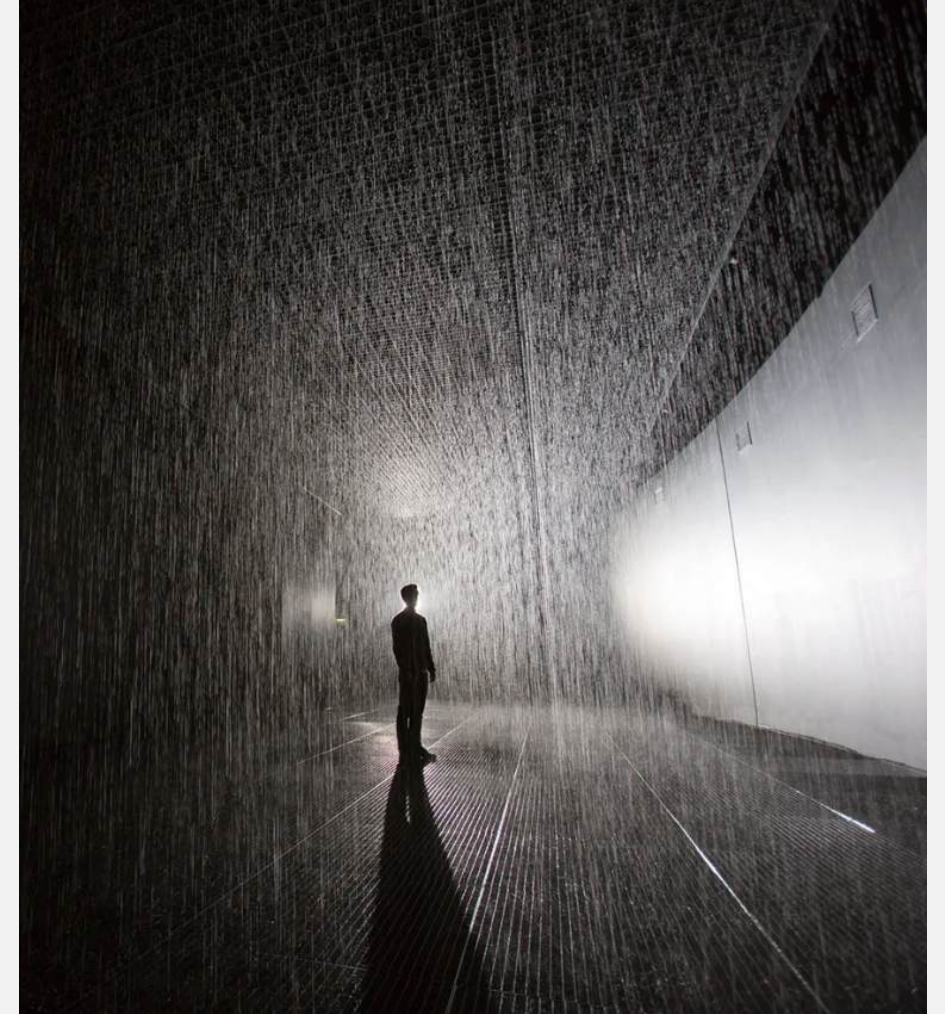
Design Philosophy in Creative Robotics

- Think about these when you are developing your own creative practice:
 - What behavior are you trying to create?
 - What do you want viewers to get out of it?
 - What are the interactions that you want to design?
 - What is the minimum sensing needed to achieve it?
 - When does [this sensor] unlock possible new opportunities?

Robotics x Machine Learning and AI

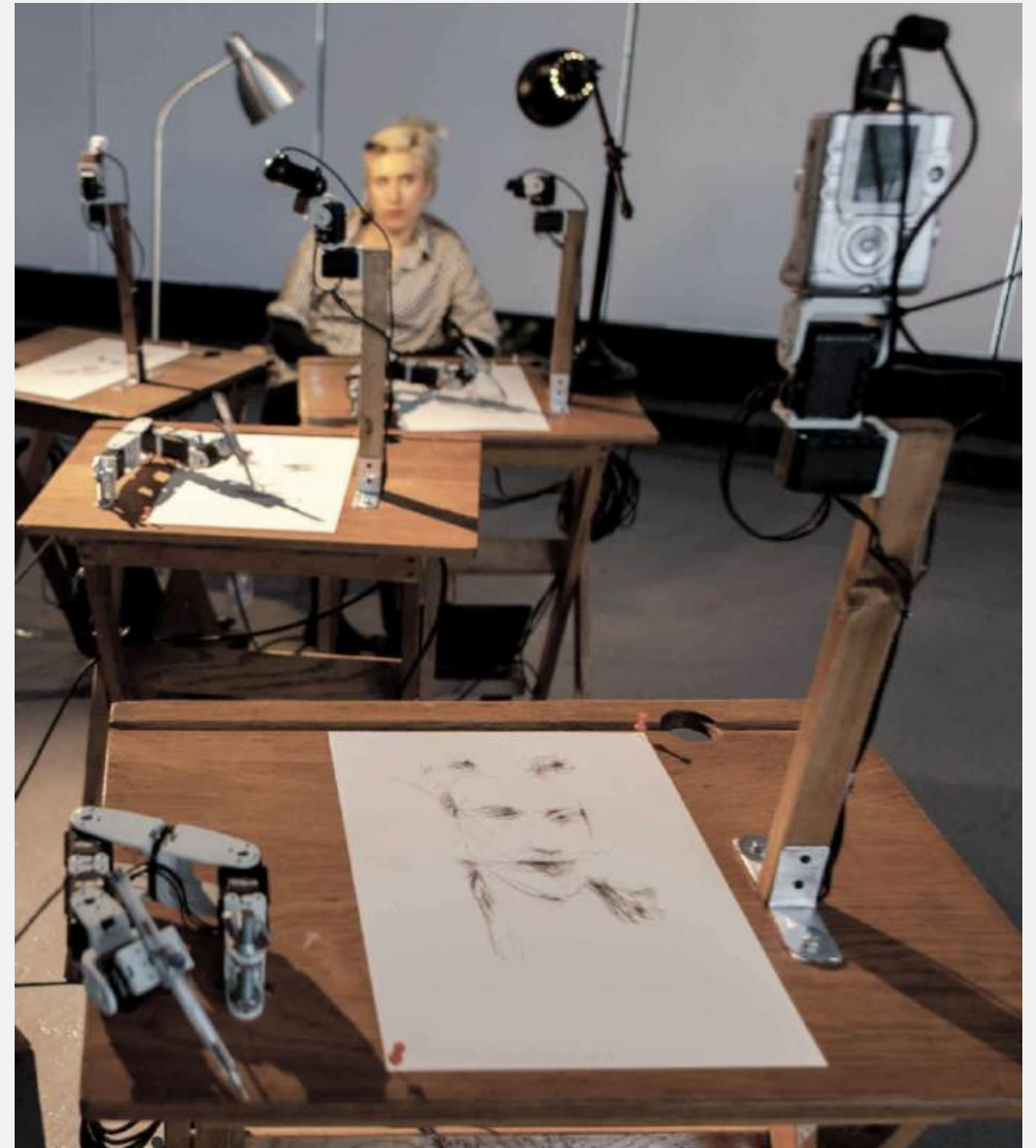
Random International: Rain Room

- Motion tracking using 3D cameras
- Rain stops where people stand/walk
- Highly immersive computer vision-based interaction
- Website: <https://www.random-international.com/rain-room>
- Relevant to: Motion detection (Week 2-3), blob tracking, background subtraction



Patrick Tresset: Paul robots

- Robot that observes and sketches human subjects
- Using computer vision to “see” the sitter
- Translates visual input to drawing output of portrait drawings in real-time
- Clear sense-plan-act model
- Website: <https://patricktresset.com/new/project/6-robots-named-paul-2012/>
- Relevant to: Object detection (Week 3-4), feature extraction, real-time processing



Tomo Kihara: How (not) to get hit by a self driving car

- A game installation that challenges people to cross a virtual street undetected by an AI-powered self-driving car simulation.
- I tricked the game by putting an umbrella in front of my whole body
- Critical perspective on computer vision, showing understanding of algorithms to subvert them
- Pedagogical value: Introduces ethics/implications of computer vision technology
- Website: <https://www.tomokihara.com/en/how-not-to-get-hit-by-a-self-driving-car.html>
- Relevant to: Face detection (Week 3-4), understanding how computer vision systems work/fail



Sougwen Chung: Drawing Operations

- Human-robot collaborative drawing
- ML trained on her drawing style
- Multiple iterations
- Computer vision + machine learning + robotics integration
- Website: <https://sougwen.com/project/drawing-operations>
- Relevant to: Neural networks (Week 4-5), training models (Week 5-6), real-time computer vision applications



Universal Everything: Future You

- Uses pose detection of computer vision
- Real-time response to human movement
- Creates unique visual avatars for participants
- Video: <https://www.youtube.com/watch?v=0PNUvwVd93Y>
- Website: <https://www.universaleverything.com/media-art/future-you>
- Relevant to: Pose detection, skeleton tracking



Computer Vision Enables New Interactions

- What computer vision makes possible that simple sensors cannot:
 - Recognize specific people, objects, or actions
 - Understand complex gestures and poses
 - Interpret facial expressions and emotions
 - Track multiple entities simultaneously
 - Understand spatial relationships and scene context
- Results: Even though computer vision is not perfect, it can still provide richer, more nuanced human-robot interactions

Today's Lab

OpenCV: Our Primary Tool

- OpenCV (Open Source Computer Vision Library)
- Open source computer vision and ML software library
- 2500+ optimized algorithms
- Classic and state-of-the-art computer vision and ML algorithms
- Can be used to:
 - Detect and recognize faces
 - Identify objects
 - Track camera movements and moving objects
 - And more!

Examples of Applications:

- Remove red eyes from flash photos
- Follow eye movements
- Recognize scenery and overlay AR markers
- Extract 3D models from images
- Stitch images for high-resolution scenes
- Find similar images in databases
- Classify human actions in videos
- Track moving objects

Today's Lab

- We have three Jupyter Notebooks today for in-class lab that you can download from the [repository](#), or fork the repository:
 - 01 — Conda and OpenCV set up
 - 02 — Python refresher
 - 03 — OpenCv “Hello World” exercise
- If you don't finish them during lab today, please complete them before the next class

Lab Time